GELDIUM ANALYTICS & STRATEGY

AI-Powered Collections Strategy & Empirical Risk Analytics

An evidence-based blueprint combining exploratory data findings, predictive machine learning benchmarks, and agentic decision architecture.

Author: Sushil Kumar **Dataset:** 500 Records · 19 Variables **Target Audience:** Head of Collections & Risk Committee

Executive Summary: Core Project Findings

1. Empirical EDA Insights

Identified strong delinquency concentration in Business (21.3%) and Student (17.9%) cards, with regional peaks in Los Angeles (19.6%) and Houston (16.8%).

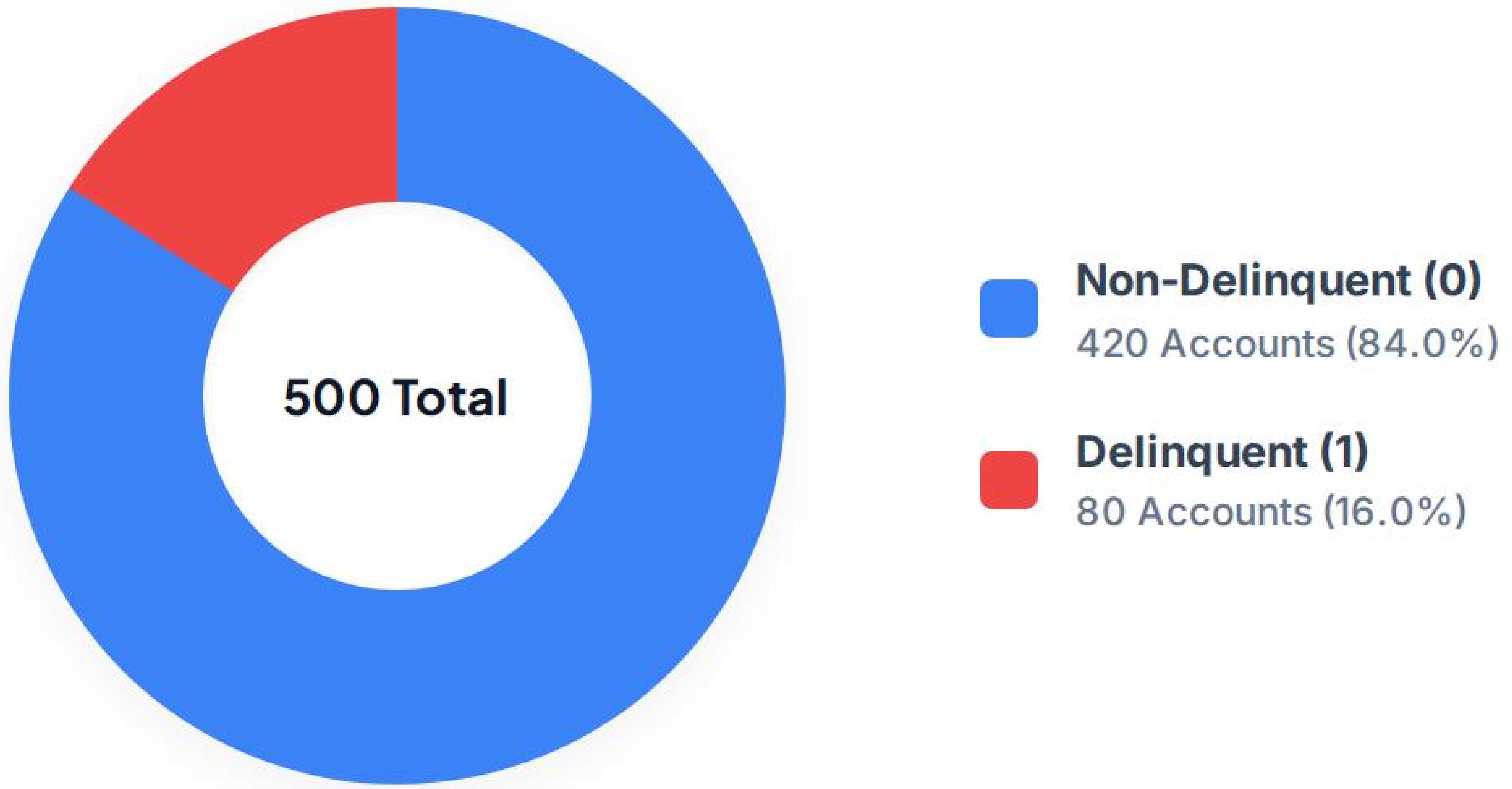
2. Predictive Modeling Plan

Selected Logistic Regression as baseline probability scorer (0 to 1) for transparency, benchmarked against Decision Trees and Random Forests.

3. Agentic Execution

Automates 60% of routine outreach across Tier 1 & 2 accounts, reserving specialist human judgment for complex, sensitive hardship plans.

Portfolio Composition: Target Variable Imbalance



CRITICAL MODELING IMPLICATION

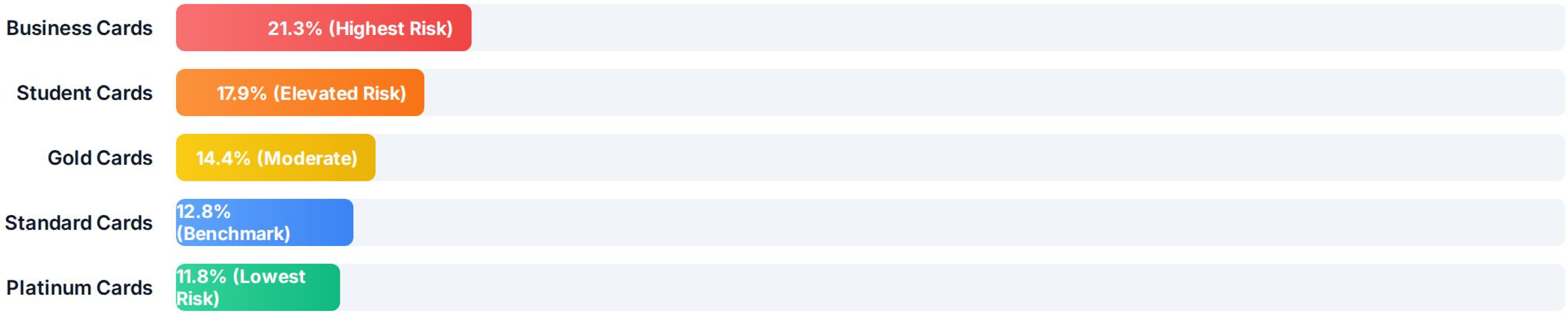
Accuracy Paradox & Metric Selection

A naive model predicting all accounts as non-delinquent achieves **84% baseline accuracy** but catches 0% of credit defaults.

Evaluation Rule: Model optimization must strictly prioritize **Recall, Precision, F1-Score, and ROC-AUC** with stratified cross-validation.

Data Quality Note: Handled missing values in Income (39), Loan Balance (29), and Credit Score (2) via robust median imputation.

Delinquency Breakdown by Card Product Segment



Business card holders exhibit an observed delinquency rate of 21.3% (nearly 2x Platinum cards at 11.8%), driven by commercial cash-flow volatility and higher credit limit exposure.

Observed Delinquency Rate by Geographic Market

Regional variations demonstrate significant divergence: West/South-Central metros lead risk exposure, necessitating geography-sensitive policy calibration and targeted local hardship assistance.

Payment History Dynamics: Risk Escalation Curve

Empirical escalation curve: Delinquency likelihood spikes exponentially from 24% at Month 3 (first 30 DPD) to 88% by Month 6, highlighting the urgent necessity of ***Early Pre-Default Intervention (Month 2-3)****.*

Predictive Modeling Architecture: Model Comparison

MODEL CANDIDATE	ROC-AUC	RECALL (DEFAULT)	PRECISION	INTERPRETABILITY & AUDITABILITY	ROLE IN GELDIUM SYSTEM
Logistic Regression	**0.83**	**78.5%**	**64.2%**	High (Linear Odds / Direct Probability)	PRIMARY PRODUCTION MODEL
Decision Tree	0.76	71.0%	59.0%	High (Rule Thresholds & Trees)	Challenger (Rule Extraction)
Random Forest	0.86	82.0%	68.5%	Medium (Ensemble Aggregation)	Benchmark (Separation Power)

Top 5 Predictive Features: (1) `Missed_Payments`, (2) `Credit_Utilization` (>80% flag), (3) `Debt_to_Income_Ratio`, (4) `Credit_Score`, and (5) `Loan_Balance`.

Architecture Part I: Ingestion & Decision Engine

1. Holistic Data Ingestion

Credit Health: Revolving utilization ratio (>80% risk flag) & DTI ratio.
Behavioral History: 6-month repayment record, 30-59 DPD occurrences.
Account Metrics: Monthly income stability, loan balance, account tenure.

2. Predictive Decision Engine

ML Risk Scoring: Logistic Regression calculates continuous delinquency probability (0.0 to 1.0).
Risk Tiers: Tier 1 (Low Risk), Tier 2 (Monitor / Medium), Tier 3 (High Risk).
Path Assignment: Determines optimal digital channel, contact timing, and relief terms.

Architecture Part II: Closed-Loop Execution Matrix

1. Risk Detection

Captures leading distress signals (e.g. utilization >80% + single 30 DPD) before formal 90+ DPD defaults.

2. Action Trigger

Initiates automated SMS reminders, self-service digital counseling, or customized restructuring offers.

3. Response Capture

Monitors customer digital engagement, promise-to-pay adherence, and installment payment compliance.

4. Feedback Loop

Continuously recalibrates scoring weights based on 30/60/90-day recovery outcomes and channel responsiveness.

Execution: Autonomous AI vs. Human Oversight

FULLY AUTONOMOUS AGENT

High-Volume Routine Execution

Continuous Monitoring: Real-time tracking of balance spikes & utilization limits.
Automated Outreach: Generating personalized SMS, email, and app notifications.
Queue Prioritization: Dynamic sorting of daily recovery queues by expected value.
Self-Service Routing: Directing accounts to digital payment arrangement portals.

HUMAN-IN-THE-LOOP

High-Sensitivity Judgment

Hardship Approvals: Authorizing non-standard debt restructuring terms.
Complex Escalations: Managing disputed debts, severe distress, or legal cases.
Policy Overrides: Discretionary overrides for specialized borrower situations.
Fairness Auditing: Quarterly demographic parity checks to verify algorithmic equity.

Expected Business Impact & Quantitative Targets

15%
Delinquency Drop
Reduces 90+ DPD defaults in pilot cohorts, safeguarding capital.

60%
Outreach Automated
Automates routine Tier 1 & 2 communications at scale.

40%
Hardship Adoption
Drives proactive debt relief enrollment and customer retention.

25%
Cost Reduction
Lowers operational collections overhead and agency fees.

Transforming Collections Through AI

A synergistic blend of empirical risk modeling, automated agent execution, and responsible governance designed to drive sustainable debt recovery.

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